An explicit constant for [OpenAI, *Uniform exclusion of Landau–Siegel zeros*] with H free: $c \geq 4.467 \times 10^{-3}$ (with Lu–Zaman–Zhao)

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Version 1 (`note.md`, `note.tex`, `note.pdf`) is kept unchanged. This PDF is built from `docs/effectivity/note-v2.tex`, which follows `note-v2.md`. Computation: `src/effectivity.py`; output: `results/2026-10-20-effectivity/effectivity-output-v2.txt`. Comparison with Banks: `src/banks_compare.py`; output: `results/2026-10-20-paper-prep-2/banks-compare-output.txt`; table: `docs/effectivity/banks-comparison-v2.md`. Derivation: `docs/effectivity/derivation.md`. Sources: [S1]–[S10], listed at the end; [S1]–[S8] are as in `docs/effectivity/sources.md`, where the quotations are; [S10] is described in `refs/banks-2610.15483.md`.

Author's note. The derivation and the Arb code were produced with Claude (Anthropic) via Claude Code. [JOE: what you checked and how]

1 Background: Banks's explicit constant

[S1] proves that some absolute $c > 0$ exists. Its proof argues by contradiction along a sequence with $q \to \infty$ and $\delta := (1-\beta)\log q \to 0$, and leaves every analytic constant implicit.

W. D. Banks [S10] (arXiv:2610.15483, dated October 11, 2026) rearranges the argument as a direct proof with explicit constants:

- $\delta_0 = 81/(5888\, e^{3/2}) = 3.0696 \cdot 10^{-3}$ for $q \geq 10^{10}$ (his Theorem 1.1 and Remark 7.1);
- with Lu–Zaman–Zhao's computation, $(1-\beta)\log q \geq 3.0696 \cdot 10^{-3}$ for every $q \geq 3$. He states this rounded, as $3.0 \cdot 10^{-3}$.

Kerr (arXiv:2610.21099) and Green (hexagon:2610.00311v2, 2026-10-17) give no explicit constant.

This note uses the same master inequality as Banks's §7, with H left free and a few other changes (§4).

2 Statement

Assuming the proof in [S1] is correct (see §8), every real zero $\beta \in (0,1)$ of every primitive nonprincipal real Dirichlet L-function of conductor $q \geq 3$ satisfies

$$(1-\beta)\log q > 4.467 \times 10^{-3} \quad \text{using Lu–Zaman–Zhao's computation for } q \leq 10^{10} \text{ (the LZZ range)},$$

$$(1-\beta)\log q > 3.399 \times 10^{-3} \quad \text{using only Platt's computation for } q \leq 4 \cdot 10^{5} \text{ (the Platt range)}.$$

The argument of [S1] alone, without any computation for small q, gives $(1-\beta)\log q > 5.04\times 10^{-4}$ for every $q \geq 3$ except $q = 8$. These bounds are certified in Arb ball arithmetic from exact rational parameters.

3 Explicit inputs

Definitions. Here $M = N^4$ and $U = N^{4/3}$, with $L = \log U$. Banks writes N^4 for our M and uses M for our $U = N^{4/3}$. The real variable of the L-function is σ throughout. derivation.md writes s for it; s is kept for the shift in `docs/effectivity/derivation-banks-device.md`.

Input	Statement used	Source and range
Hadamard identity (2.3)	$\mathrm{Re}(-L'/L)(\sigma,\chi) = \frac{1}{2}\log(q/\pi) + \frac{1}{2}\psi((\sigma+\varepsilon)/2) - \sum_\rho \mathrm{Re}\,\frac{1}{\sigma-\rho}$, terms ≥ 0, real $\sigma > 1$	as stated in [S1] §2; checked against Davenport, 2nd ed., Ch. 12 (17)–(18), p. 83
ζ log-derivative	$-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$, $\sigma > 1$	[S6] Lemma 12; final elementary inequality proved here for all $\sigma > 1$ (§4)
Digamma	$\psi((\sigma+\varepsilon)/2) \leq \psi(3/2) = 2 - \gamma - 2\log 2$ for $1 < \sigma \leq 2$	ψ increasing
Mertens	$\log x + E - \frac{1}{2\log x} < \sum_{p\leq x}\frac{\log p}{p}$ $(x > 1)$; $\sum_{p\leq x}\frac{\log p}{p} < \log x + E + \frac{1}{2\log x}$ $(x \geq 319)$	Rosser–Schoenfeld Thm 6, (3.21)–(3.22), p. 70, read in the original
Chebyshev	$\theta(x) < (1 + 7.5\cdot 10^{-7})x$, $x > 0$	[S5] Platt–Trudgian Cor. 2
Primes dividing q	$\sum_{p\mid q}\frac{\log p}{p} \leq \min\left(\frac{1}{2}, \frac{\log 2}{2\log q} + \frac{1}{3}\right)\log q$	elementary
Lemma 6	$S_2 \leq 192\,MH^{-1/3}U$, $S_1 \geq MH^{2/3}U/(4\cdot 97^2)$, given $N \geq H$ and $H^{2/3}N^{4/3} \geq 2\cdot 97^2$	[S1] lines 517–550 (= Banks Lemma 4.2)
Small q	no real zero with $\sigma > 0$ for $q \leq 4\cdot 10^5$ ([S9]); $\delta > 1/5$ for $q \leq 10^{10}$	[S9] Thms 10.1–10.2, read in the original; [S2] Theorem 1.1

4 The master inequality, the parameters, and the comparison with Banks

Combining the explicit Lemma 2, (5.1), (5.6) with the θ bound, and Lemma 6 (derivation.md §§1–5): if a real zero with $\delta = (1-\beta)\log q$ exists and $N \geq H$, then

$$1 \leq \tfrac{3}{4}(1+\rho_H) + \frac{C_\ell + \frac{1}{2}(1+\rho_H)}{\lambda} + \frac{e}{2}\,\delta\lambda + \frac{C_H + (1+\rho_H)\log 8}{L} + \frac{(1+7.5\cdot 10^{-7})\cdot 4\cdot 97^2}{H^{2/3}L} + \frac{6\cdot 97^2}{H^{2/3}U},$$

where $\rho_H = 7{,}226{,}112/H$, $C_\ell = e/4 + c_q$, $C_H = \log H + \frac{1}{2\log H} + \frac{1}{2L} + \frac{e}{2}\kappa$, $\kappa = \frac{1}{2}\psi(3/2) - \frac{1}{2}\log\pi = -0.55412\ldots < 0$, $L = \log U = \frac{4}{3}\log N$ and $\lambda = L/\log q$. This is Banks's Step 3 inequality (§7), term for term, in a different accounting (`banks-comparison-v2.md`, rows 1–21).

Take $N = \lceil q^\gamma \rceil$. For all $q \geq q_0$:

- $\lambda \in [4\gamma/3,\ 4\gamma/3 + \frac{4}{3}\log(1+q_0^{-\gamma})/\log q_0]$;
- every term except the δ term is largest at $q = q_0$;
- so every real zero with $q \geq q_0$ has $\delta > \delta_0(q_0,\gamma,H)$, with δ_0 given in closed form (function `delta0` in `src/effectivity.py`).

For fixed (γ, H), δ_0 is determined exactly; no fixed split of the margin is used. The parameters were found by a floating-point search, then certified in Arb at the exact values below (every printed bound is the Arb lower endpoint rounded down).

Range	q_0	γ	H	c_q	certified $\delta_0 >$
Platt	400,001	201/10	1,995,262,314	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{3.3992904026 \times 10^{-3}}$
Lu–Zaman–Zhao	$10^{10}+1$	153/10	2,818,382,931	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{4.4673780338 \times 10^{-3}}$
argument alone	3	671/5	1,122,018,454	$\frac{1}{2}$	$\mathbf{5.0425352961 \times 10^{-4}}$

Unlike Banks's §7, our rows do not need $H \le q$: the Platt row has $H > q_0$. Banks uses $H \le q$ only to get $N \ge H$, which γ secures here (side condition $N \ge H$ below).

From Banks's $3.0696 \cdot 10^{-3}$ to $4.467 \cdot 10^{-3}$ (COMPUTED; `banks-comparison-v2.md` (c)). The waterfall below starts from Banks's published value at $q_0 = 10^{10}$ and switches his choices to ours one at a time. γ is re-optimized at each step; H is the paper's until the last step. The factors depend on the order.

Change from Banks's choice to ours	Factor
exact solve instead of the 3/32 split and the $\log q > 23$ absorption	$\times 1.024$
tight upper end of λ ($N \le q^\gamma + 1$, not $q^{\omega+1}$)	$\times 1.046$
our S_1, S_2 bounds ($S_1 \ge N^4 H^{2/3} U/(4 \cdot 97^2)$, not $N^4 U$)	$\times 1.044$
Platt–Trudgian θ bound instead of $\theta(M) \le 2M$	$\times 1.003$
$\kappa < 0$ kept (with the Mertens terms)	$\times 1.02$
shift 1 in $\sigma = 1 + s/\log X$, instead of Banks's 3/2 (a cost)	$\times 0.88$
our primes-dividing-q bound, instead of Banks's $\frac{1}{2}S_0$ device (a cost)	$\times 0.86$
H free ($\gamma = 153/10$, $H = 2{,}818{,}382{,}931$), instead of $H = 86{,}713{,}344$	$\times 1.68$

Check (COMPUTED, `results/2026-10-20-paper-prep-3/note-v2-factor-check.txt`): the exact factors multiply to the ratio $4.4673780338/3.0695555319 = 1.45538$. The seven rounded factors other than θ multiply to 1.45018 (-0.36%), and with θ to 1.45453 (-0.06%); both lie within the rounding of the stated factors. At the paper's own $H = 86{,}713{,}344$, the same three ranges give 2.064, 2.652 and 0.3245 ($\times 10^{-3}$). A further refinement of the accounting is under review and is not used here.

Side conditions (derivation.md §7), certified at q_0 and implied for all $q \ge q_0$: $H \ge 319$; $N \ge H$ ($\gamma \log q_0 > \log H$); $H^{2/3} N^{4/3} \ge 2 \cdot 97^2$ (from $N \ge H \ge 319$); $U \ge 3$, and $\sigma = 1 + 1/L \in (1, 2]$; $\kappa < 0$.

Lemma 12's last step. $\varphi(\sigma) = \frac{1}{2\sigma - 1} - \left[\frac{1}{\sigma - 1} - \frac{1 - 2^{-\sigma}}{\sigma} - \frac{3^{1-\sigma}}{\sigma - 1}\right] > 0$ for all $\sigma > 1$: on $(1, 1.01]$ by the exact chain $(1 - 3^{-t})/t \le \log 3$, $(1 - 2^{-\sigma})/\sigma \ge 1/(2\sigma)$, reducing it to $\log 3 - 1/2.02 < 1/1.02$; on $[1.01, 5]$ by Arb on 4,000 intervals (minimum certified lower bound of φ: 0.03878); for $\sigma \ge 5$ by $\varphi > 0 \Leftarrow \sigma^2 - 5\sigma + 2 > 0$.

5 Remarks

- **Banks's numbers recomputed.** Each of the 12 numeric claims in Banks's Lemma 2.1 and §7 was recomputed independently in Arb, and all 12 hold. His $\delta_0 = 81/(5888e^{3/2})$ recomputes as $[0.00306955553193322 \pm 2.9 \cdot 10^{-18}]$ (`src/banks_compare.py`, Part A).
- **Lemma 3.** Our Lean proof of [S1]'s Lemma 3 (`lean-checks/Lemma3.lean`, theorem `Lemma3.interpolation`) has hypotheses identical to Banks's condition (3.1), the hypothesis of his Proposition 3.4 (clause-by-clause comparison: `docs/packet-related-work-draft-v2.md`, which points to the table in `docs/packet-related-work-draft.md`).

6 Small conductors

- **Platt range:** for $q \le 4 \cdot 10^5$ there is no real zero with $\sigma > 0$ at all, so the bound holds vacuously. This range includes $q = 8$, the conductor the paper's field construction excludes ($d = 2$). $q > 4 \cdot 10^5$ is covered by the first row.
- **Lu–Zaman–Zhao range:** for $q \le 10^{10}$, $\delta > 1/5$, which also covers $q = 8$; $q > 10^{10}$ is covered by the second row; so $c = \min(4.467 \cdot 10^{-3}, 1/5) = 4.467 \cdot 10^{-3}$.
- **Argument alone:** covers $q \ge 3$ with $q \ne 8$ by taking γ large ($\lambda \approx 179$), i.e. $N \approx q^{134}$. $q = 8$ needs a separate input.

7 The final constant

$$c = 4.467 \times 10^{-3} \text{ (with Lu–Zaman–Zhao)}, \qquad c = 3.399 \times 10^{-3} \text{ (with Platt only)},$$

both rounded down from the certified values.

Context.

- The companion paper on the zero-free half-plane $\operatorname{Re} s > 7/8$ would, if correct, give $1 - \beta \ge 1/8$ for every such β, a far stronger statement.
- A real zero with $\delta \approx 0.003$ is far outside anything ever observed. These numbers are for checking the argument, not a claim about zeros.

Comparison with an earlier version (docs/explicit-constant.md, written 2026-10-08, public in reviewer packet v1 on 2026-10-09: tag reviewer-packet-v1, commit 3a36016, Zenodo 10.5281/zenodo.23269528). Its figures were $3.0379 \cdot 10^{-4}$ (row "all $q \ge 3$"), $1.99806 \cdot 10^{-3}$ ($q > 4 \cdot 10^5$) and $2.58930 \cdot 10^{-3}$ ($q > 10^{10}$):

- **Same inputs reproduce it.** With its inputs switched back on, this code gives slightly larger maxima ($3.040 \cdot 10^{-4}$, $2.0000 \cdot 10^{-3}$, $2.5918 \cdot 10^{-3}$, rounded down). The earlier note reported a δ with a positive safety margin ($F \approx 9.4 \cdot 10^{-5}$) rather than the maximum.
- **"All $q \ge 3$" is used there twice:** its table row "all $q \ge 3$" ($3.0379 \cdot 10^{-4}$) covers small q the same way as row 3 here: by taking λ large, not by any computation ($q = 8$ again excepted). Its bold conditional statement "$(1 - \beta) \log q \ge 0.00258 \ldots$ with $q \ge 3$" combines the $q > 10^{10}$ row with the small-q computations of Lu–Zaman–Zhao (and Platt). Source: `results/2026-10-20-paper-prep-4/task2-earlier-constant.txt`.
- **It used three inputs replaced here:** an unsourced "$\sum_{p \le x} \log p/p < \log x$" for the primes up to H, which costs $|E| = 1.33$ in C_H against the sourced Rosser–Schoenfeld upper bound; Rosser–Schoenfeld Thm 9 for θ; a fixed H.

- **The gain comes mainly from freeing** H ($\times 1.6$). Of the other changes, $c_q = \frac{1}{2}$ is worse than its c_q at large q but better at $q = 3$, and the sourced Mertens bound and the θ bound are improvements.

8 Assumptions (everything the constant depends on)

1. **The paper's proof is correct:** in particular Lemma 3, Corollary 4, Lemma 6, Lemma 7 and the bounds (5.1) and (5.6).
   - This note does not re-verify them mathematically.
   - Independent evidence: the Lean formalization of Theorem 1 is accepted by the official comparator with two kernels (Lean and nanoda), including a version rerouted through the paper's own Lemma 3. See the replay packet (DOI 10.5281/zenodo.23269527) and the statement audit on `grwtsk/openai-math` #40.
   - The Lean proof has existential constants only and does not contain these numbers.
2. **The Hadamard identity (2.3).** Checked against Davenport, *Multiplicative Number Theory*, 2nd ed., Ch. 12, (17) and (18) on p. 83, with $a = \varepsilon$ (p. 70). Taking real parts of (17) and using (18), $\operatorname{Re} B(\chi) = -\sum \operatorname{Re} 1/\rho$, gives (2.3).
3. **Rosser–Schoenfeld Theorem 6,** read in the original: (3.21) for $1 < x$ and (3.22) for $319 \le x$, p. 70. $E \approx -1.33258\,22757\,33221$, (2.11). This agrees with the quotation in arXiv:1703.08032 and with the enclosure in `src/effectivity.py`.
4. **Platt (2016) [S9],** read in the original. Theorem 10.1 states GRH for primitive characters of modulus $q \le 400\,000$, up to a positive height. Theorem 10.2 states $L(1/2, \chi) \ne 0$ for $q \le 2\,000\,000$. Together (our inference) they rule out a real zero in $(0, 1)$ for $q \le 4 \cdot 10^5$, matching the Lu–Zaman–Zhao summary. Needed only for the $3.399 \cdot 10^{-3}$ statement.
5. **Lu–Zaman–Zhao Theorem 1.1** (arXiv:2602.03626v1, a preprint), read in the original. Needed only for the $4.467 \cdot 10^{-3}$ statement.
6. **Platt–Trudgian Corollary 2,** read in the original.
7. **Lemma 12 of arXiv:2511.13898** (a preprint), read in the original. Its reduction steps are taken from that proof; its final inequality is proved here.
8. $\chi(p) = (d/p)$ **for odd** $p \nmid q$**, and** $|d| \le q$**,** as stated in the paper.

Not used: any unproved hypothesis (GRH or similar), or any floating-point value in a claimed inequality.

References

[S1] OpenAI, *Uniform exclusion of Landau–Siegel zeros*, October 1, 2026, `github.com/openai/math`, commit adc7f12, family 003.

[S2] R. F. Lu, A. Zaman, H. Zhao, *Numerical computations concerning Landau–Siegel zeros*, arXiv:2602.03626v1 (2026).

[S3] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. 6 (1962), 64–94, `doi:10.1215/ijm/1255631807`.

[S4] C. Axler, *New estimates for some functions defined over primes*, arXiv:1703.08032.

[S5] Platt, Trudgian, *On the first sign change of* $\theta(x) - x$, arXiv:1407.1914.

[S6] T. Zhao, *Upper bounds on gaps between zeros of L-functions*, arXiv:2511.13898, Appendix, Lemma 12.

[S7] H. Davenport, *Multiplicative Number Theory*, 2nd ed., rev. H. L. Montgomery, Springer GTM 74.

[S8] Elementary facts (no citation): ψ increasing on $(0,\infty)$, $\psi(3/2) = 2-\gamma-2\log 2$, and $\log p/p \le (\log p)/2$ for $p \ge 2$ (`docs/effectivity/sources.md`, S8).

[S9] D. J. Platt, *Numerical computations concerning the GRH*, Math. Comp. 85 (2016), 3009–3027, `doi:10.1090/mcom/3077`.

[S10] W. D. Banks, *On the OpenAI proof that Landau–Siegel zeros do not exist*, arXiv:2610.15483v1 (2026).